

Changelog — The UAT Physical Guarantee of Navier-Stokes Regularity

All notable changes to this project will be documented in this file.

Timestamps are immutable via Git commit history — proving priority 11 months before OpenAI Sep 8, 2026.

GitHub:

https://github.com/miguelpercu/The_UAT_Physical_Guarantee_of_Navier_Stokes_Regularity

Author: Miguel Angel Percudani — ORCID 0009-0007-1748-3212

Related DOIs: 10.5281/zenodo.17729221 (UAT) | 10.5281/zenodo.18210808 (UPC) |

10.5281/zenodo.22678424 (Antifrequency)

[v1.0] - 2025-10-26 — Original Physical Guarantee (Prior Art)

First public release — 11 months BEFORE OpenAI blow-up claim (Sep 8, 2026)

Added

- `Navier_Stokes.pdf` (5 pages, LaTeX) — The UAT Physical Guarantee of Navier-Stokes Regularity
 - Theorem: $\rho_{\max} = M_{\text{Planck}} / L_{\text{Planck}}^3 \approx 5.155e96 \text{ kg/m}^3$
 - Bound: $\epsilon_{\text{kin}} = 1/2 \rho_{\text{fluid}} |v|^2 \leq \rho_{\max} c^2$
 - Result: $|v|_{\max} \text{ finite} \Rightarrow \text{smooth}$
- `limitación_del_caos_de_Navier_Stokes.py` — Python implementation (v1 code)
 - Output v1 (without c^2 in code): $|v|_{\max} = 1.015e47 \text{ m/s}$ finite
 - Note: LaTeX Eq.(5) includes c^2 , code v1 omitted it. Both finite, corrected in v2.
- Physical interpretation: mathematical singularities require infinite energy density, violating LQG discreteness.
- UAT Guarantee Theorem by contradiction.

Status

- This version establishes **physical guarantee**, NOT formal Clay Millennium proof (pure PDE). Explicitly stated in abstract.

Verification

- Commit hash (GitHub history): preserved as first commit 2025-10-26
 - Zenodo v1 (when restored): original upload date Oct 2025
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[v2.0 / v9.1] - 2026-09-09 — Addenda: Physical Regularization vs Mathematical Blow-up (Response to OpenAI)

Released 1 day AFTER OpenAI announcement Sep 8, 2026

Context

- OpenAI announced solution claiming finite-time blow-up, Cases C/D Clay, 1000 agents Euler 50h + 10,000 agents Navier-Stokes 11h, total 88h, 130B tokens, ~\$15M, Lean 17h verification. Mechanism: vortical filament spiraling inward + axial stretching, total energy finite but local density $\rightarrow \infty$. Controversy Buckmaster/Alpöge. Clay President Martin Bridson: evaluation deliberately unhurried.

Added

- `Addenda_v9_1_NavierStokes_Rgeom.pdf` (1 page) — Navier-Stokes Millennium Blow-up as Geometric Filter Failure, Parallel to $R_{geom}=0.2791$ Regularization
 - Links blow-up to UAT regularization: $k_{early}=0.967$, $\Delta\theta=43.515^\circ$, $Cycle=348.12^\circ$, deficit $11.88^\circ=3.3\%$ epsilon, $R_{geom}=0.279182$ (4.7x smaller than random $p<0.001$), $2*R_{geom}=0.558364$
 - Thesis: Pure NS $\rightarrow$ blow-up possible (OpenAI), UAT-regularized NS with $V_{min}=L_P^3 \rightarrow |v|_{max}$ finite $\rightarrow$ smooth
- `README_UAT_NavierStokes.md` — Expanded README with full comparison table OpenAI vs UAT
 - Clarification added: "Esto NO es una prueba formal del Milenio de Clay (PDE pura). Esta es la garantía física de que tal regularidad debe cumplirse en la naturaleza"

debido al límite $\rho_{\max}$ de LQG, es decir, la regularización física que impide la explosión matemática recientemente afirmada por OpenAI (8 Sep 2026). Navier-Stokes puro $\rightarrow$ posible explosión (OpenAI C/D) / Navier-Stokes regularizado por UAT con $V_{\min}=L_P^3 \rightarrow |v|_{\max}$ finito $\sim 1e47$ - $1e56$ m/s $\rightarrow$ suave."

- `Note_03_08_2026.pdf` — Before the Metrics: Observer's Dilemma (fly 200-250Hz vs human 50-60Hz filter origin of question "what is real time?")
- `supplementary_script.py` — Simulation human 50Hz blurred >70% ambiguity vs fly 200Hz discrete <20% deterministic, 1000Hz quantum reality
- Corrected calculation in README: with $c^2 \Rightarrow \sim 9.6e56$ m/s finite (physically correct), without $c^2 \Rightarrow 1.015e47$ m/s (v1 output)
- Zenodo description updated for v2: same concept DOI, new version DOI, title "The UAT Physical Guarantee... From Planck-Scale Finite Bound to $R_{\text{geom}}=0.2791$ Regularization — Addenda v9.1 in Response to OpenAI Millennium Blow-up Claim (Sep 2026)"

Changed

- README now explicitly distinguishes physical guarantee vs formal Clay proof to avoid misinterpretation post-OpenAI.
- No previous files modified in v1.0 history — only additions via new commit to preserve immutable timestamps for priority.

Priority Proof

- GitHub commit history shows v1.0 (Oct 2025) exists 11 months before OpenAI (Sep 2026)
- This CHANGELOG provides official timeline for Clay Institute, Zenodo reviewers, and academic citations

[Unreleased] — Roadmap

- Formalize $v_{\text{eff}}(l) = v * (1 + (L_P/l)^\alpha)$ modified viscosity
- Ψ -NSE v1.0 term: $-\gamma_c \Psi(t)$ u Schrödinger-Poisson phase field regularization (see Zenodo 2026-02-27)
- Lean 4 formalization of $\rho_{\max}$ bound as axiom for physically regularized NS

How to tag releases (for maintainer)

bash

```
1 # Already done for v1.0
1 git tag -a v1.0 -m "Original physical guarantee Oct 26 2025, prior art 11
1
1 # Now for v2.0 / v9.1
1 git add Addenda_v9_1_NavierStokes_Rgeom.pdf README_UAT_NavierStokes.md CHA
1 git commit -m "Add v9.1 addenda response to OpenAI blow-up Sep 8 2026, lin
1 git tag -a v2.0 -m "Addenda v9.1 Sep 9 2026 response to OpenAI blow-up, R_
1 git tag -a v9.1 -m "Same as v2.0, linked to UAT Theory of Desynchronized T
1 git push origin main --tags
```

Citation

Percudani, M.A. (2025–2026). The UAT Physical Guarantee of Navier-Stokes Regularity: From Planck-Scale Finite Bound to R_{geom} Regularization — Addenda v9.1 in Response to OpenAI Millennium Blow-up Claim. GitHub:

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